Fark denklemi ile tanımlanan sistemler

Bankadaki tasarruf hesabı örneği:

$$y[n] = y[n-1] + y[n-1] \times 0.1 + x[n]$$

$$y[n] = 1.1y[n-1] + x[n]$$

sabit-katsayılı fark denklemi (SKFD)

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

İŞARET VE SİSTEMLERİN FREKANS UZAYI ANALİZİ

Frekans yanıtı

Doğrusal zamanla değişmez bir sisteme $x[n] = e^{j\Omega n}$

girişi uygulandığı taktirde

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=-\infty}^{\infty} h[k]e^{j\Omega(n-k)} = e^{j\Omega n}\left(\sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k}\right)$$

$$H(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k}$$

$$x[n] = e^{j\Omega n} \longrightarrow y[n] = H(e^{j\Omega})e^{j\Omega n}$$

Frekans yanıtının gösterimi

$$H(e^{j\Omega}) = H_G(e^{j\Omega}) + jH_S(e^{j\Omega})$$

$$H(e^{j\Omega}) = \left|H(e^{j\Omega})\right| e^{j\angle H(e^{j\Omega})}$$

Örnek

- Bir sistem $y[n] = x[n - n_0]$ şeklinde tanımlanmıştır.

a) Sistemin frekans yanıtını bulalım.
b) Sistemin frekans yanıtını kullanarak sistemin $x[n] = \sin(\pi\, n/10)$ girişine verdiği çıkışı bulalım.

Ayrık Zamanlı Fourier Dönüşümü (AZFD)

AZFD:

$$X\left(e^{j\Omega}\right)=\sum_{n=-\infty}^{\infty}x[n]e^{-j\Omega n}$$

Ters-AZFD:

$$x[n]=\frac{1}{2\pi}\int_{-\pi}^{\pi}X\left(e^{j\Omega}\right)e^{j\Omega n}d\Omega$$

AZFD'nin periyodikliği

$$X\left(e^{j(\Omega+2\pi r)}\right)=\sum_{n=-\infty}^{\infty}x[n]e^{-j(\Omega+2\pi r)n}=\sum_{n=-\infty}^{\infty}x[n]e^{-j\Omega n}e^{-j2\pi rn}=\sum_{n=-\infty}^{\infty}x[n]e^{-j\Omega n}=X\left(e^{j\Omega}\right)$$

Genlik/Faz spektrumu

$$X\left(e^{j\Omega}\right) = \left|X\left(e^{j\Omega}\right)\right| e^{j\angle X\left(e^{j\Omega}\right)}$$

$$\left|X\left(e^{j\Omega}\right)\right| \qquad \angle X\left(e^{j\Omega}\right)$$

örnek

- Birim dürtü işareti için AZFD?

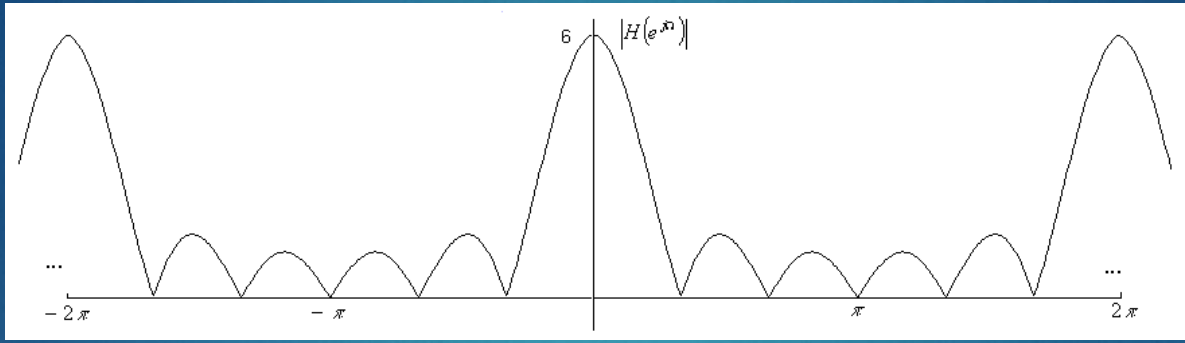
örnek

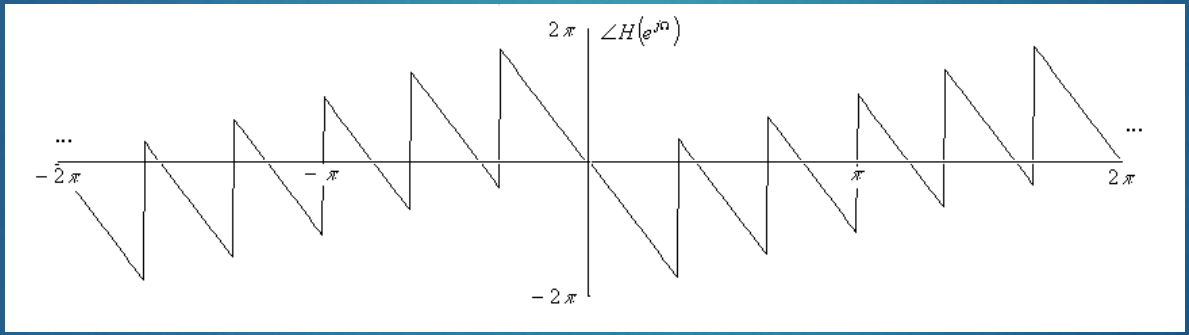
$$h[n] = \begin{cases} 1 \ , & 0 \le n \le N-1 \\ 0 \ , & \text{aksi taktirde} \end{cases} \Longrightarrow H(e^{j\Omega}) = ?$$

$$H\left(e^{j\Omega}\right) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\Omega n} = \sum_{n=0}^{N-1} e^{-j\Omega n} = \sum_{n=0}^{N-1} \left(e^{-j\Omega}\right)^n$$

$$H\left(e^{j\Omega}\right) = \sum_{n=0}^{N-1} \left(e^{-j\Omega}\right)^n = \frac{e^{-j\Omega N} - e^{-j\Omega 0}}{e^{-j\Omega} - 1} = \frac{e^{-j\Omega N} - 1}{e^{-j\Omega} - 1} = \frac{1 - e^{-j\Omega N}}{1 - e^{-j\Omega}}$$

örnek





örnek

$$x[n] = a^n u[n] \Longrightarrow H(e^{j\Omega}) = ?$$

$$X\left(e^{j\Omega}\right) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} = \sum_{n=0}^{\infty} a^n e^{-j\Omega n} = \sum_{n=0}^{\infty} \left(a e^{-j\Omega}\right)^n$$

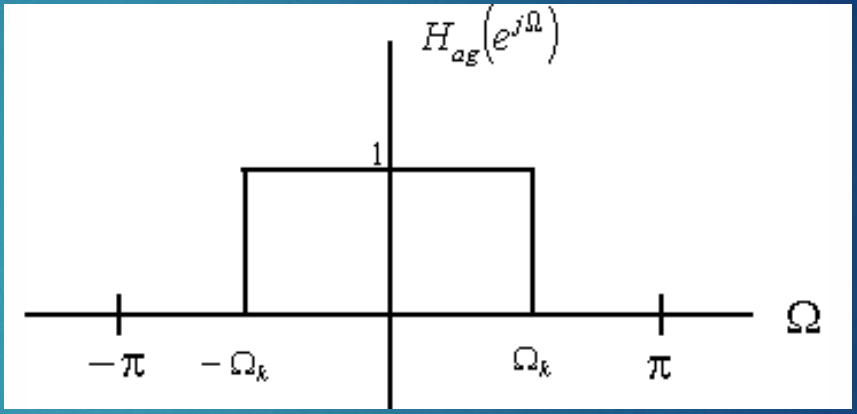
$$X\left(e^{j\Omega}\right) = \sum_{n=0}^{\infty} \left(a e^{-j\Omega}\right)^n = \frac{0 - \left(a e^{-j\Omega}\right)^0}{a e^{-j\Omega} - 1} = \frac{-1}{a e^{-j\Omega} - 1} = \frac{1}{1 - a e^{-j\Omega}} \ , \ |a| < 1$$

Örnek- İdeal AGS

- İdeal alçak-geçiren süzgeç:

$$H\left(e^{j\Omega}\right)=\begin{cases}1 \ , & |\Omega|<\Omega_k \\ 0 \ , & \Omega_k<|\Omega|<\pi\end{cases}$$

$$h[n]=?$$

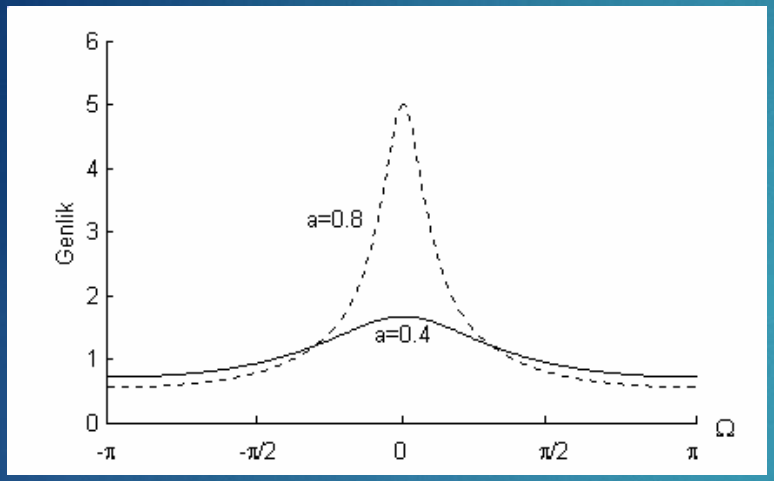


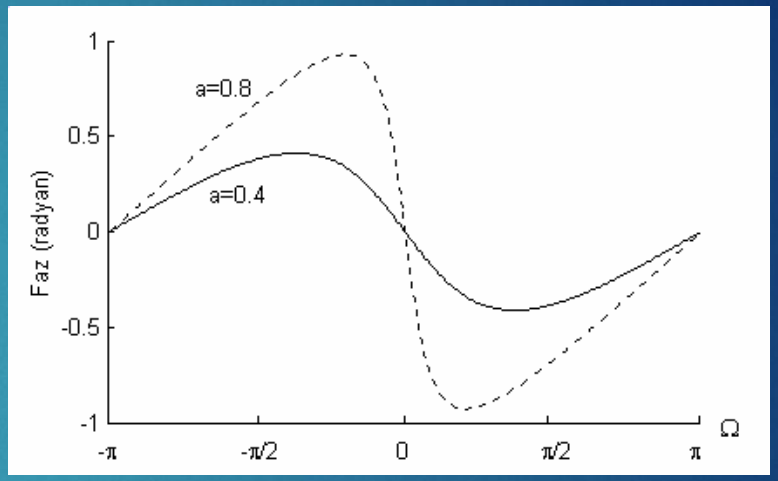
AZFD simetri özellikleri

Dizi $x[n]$	Fourier dönüşümü $X\left(e^{j\Omega}\right)$
$x^*[n]$	$X^*\left(e^{-j\Omega}\right)$
$x^*[-n]$	$X^*\left(e^{j\Omega}\right)$
Gerçek { $x[n]$ }= $x_G[n]$	$X_ç\left(e^{-j\Omega}\right)$
j Sanal { $x[n]$ }=j $x_S[n]$	$X_t\left(e^{-j\Omega}\right)$
$x_ç[n]$	Gerçek { $X\left(e^{j\Omega}\right)$}= $X_G\left(e^{j\Omega}\right)$
$x_t[n]$	j Sanal { $X\left(e^{j\Omega}\right)$}= $X_S\left(e^{j\Omega}\right)$
Herhangi gerçek bir $x[n]$	$X\left(e^{j\Omega}\right)=X^*\left(e^{-j\Omega}\right)$
Herhangi gerçek bir $x[n]$	$X_G\left(e^{j\Omega}\right)=X_G\left(e^{-j\Omega}\right)$
Herhangi gerçek bir $x[n]$	$X_S\left(e^{j\Omega}\right)=-X_S\left(e^{-j\Omega}\right)$
Herhangi gerçek bir $x[n]$	$\left\|X\left(e^{j\Omega}\right)\right\|=\left\|X\left(e^{-j\Omega}\right)\right\|$
Herhangi gerçek bir $x[n]$	$\angle X\left(e^{j\Omega}\right)=-\angle X\left(e^{-j\Omega}\right)$

örnek

$$x[n] = a^n u[n] \Longrightarrow X\left(e^{j\Omega}\right) = \frac{1}{1 - ae^{-j\Omega}}, |a| < 1$$





AZFD Teoremleri

Doğrusallık

$$x_1[n] \overset{AZFD}{\Leftrightarrow} X_1\left(e^{j\Omega}\right) \qquad x_2[n] \overset{AZFD}{\Leftrightarrow} X_2\left(e^{j\Omega}\right)$$

$$ax_1[n] + bx_2[n] \overset{AZFD}{\Leftrightarrow} aX_1\left(e^{j\Omega}\right) + bX_2\left(e^{j\Omega}\right)$$

$$x[n] = \left(0.2^n + 0.4^n\right)u[n] \Longrightarrow X(e^{j\Omega}) = ?$$

Zamanda öteleme

$$x[n] \overset{AZFD}{\Leftrightarrow} X\left(e^{j\Omega}\right) \longrightarrow x[n-n_0] \overset{AZFD}{\Leftrightarrow} e^{-j\Omega n_0} X\left(e^{j\Omega}\right)$$

$$x[n] = \left(0.2^n\right) u[n-2] \Longrightarrow X(e^{j\Omega}) = ?$$

Frekansta öteleme

$$x[n] \overset{AZFD}{\Leftrightarrow} X\left(e^{j\Omega}\right) \longrightarrow e^{j\Omega_0 n} x[n] \overset{AZFD}{\Leftrightarrow} X\left(e^{j(\Omega-\Omega_0)}\right)$$

$$x[n] = \cos\left(\frac{\pi n}{10}\right) 0.2^n u[n] \Longrightarrow X(e^{j\Omega}) = ?$$

Zamanda ters çevirme

$$x[n] \overset{AZFD}{\Leftrightarrow} X\left(e^{j\Omega}\right) \longrightarrow x[-n] \overset{AZFD}{\Leftrightarrow} X\left(e^{-j\Omega}\right)$$

$$x[n] = 2^n u[-n] \Longrightarrow X(e^{j\Omega}) = ?$$

Konvolüsyon teoremi

$$x[n] \overset{AZFD}{\Leftrightarrow} X\left(e^{j\Omega}\right) \qquad h[n] \overset{AZFD}{\Leftrightarrow} H\left(e^{j\Omega}\right)$$

$$y[n] = x[n] * h[n] \overset{AZFD}{\Leftrightarrow} Y\left(e^{j\Omega}\right) = X\left(e^{j\Omega}\right) H\left(e^{j\Omega}\right)$$

$$x[n] = 0.2^n u[n]$$

$$h[n] = \delta(n-2) + \delta(n-1)$$

$$\Longrightarrow \quad Y(e^{j\Omega}) = ?$$

Konvolüsyon teoremi

$$x[n] \overset{AZFD}{\Leftrightarrow} X\left(e^{j\Omega}\right) \qquad h[n] \overset{AZFD}{\Leftrightarrow} H\left(e^{j\Omega}\right)$$

$$y[n] = x[n] * h[n] \overset{AZFD}{\Leftrightarrow} Y\left(e^{j\Omega}\right) = X\left(e^{j\Omega}\right) H\left(e^{j\Omega}\right)$$

$$x[n] = 0.2^n u[n]$$

$$h[n] = \delta(n-2) + \delta(n-1)$$

$$\Longrightarrow \quad Y(e^{j\Omega}) = ?$$

örnek

$$x[n] = 0.2^{n} u[n-4] + 0.4^{n-1} u[n] \Longrightarrow X(e^{j\Omega}) = ?$$

örnek

$$y[n]-0.2y[n-1]=0.5x[n]-0.3x[n-1]+0.1x[n-2]$$

$$h[n]=?$$

Ayrık Fourier Dönüşümü

AFD ?

- Ayrık-Zamanlı Fourier Dönüşümü: sonsuz toplam
- Ayrık-Zamanlı Fourier Dönüşümü: frekansta sürekli bir fonksiyon

Fourier dönüşümü hesabı

- N adet frekans için hesapla. Frekans aralıkları: $\frac{2\pi}{N}$

- Toplam N adet frekans: $\frac{2\pi k}{N}, k = 0,1,\ldots,N-1$

- Fourier dönüşümü hesabı:

$$X\left(e^{j\Omega}\right)\Big|_{\Omega=2\pi k/N} = X[k], k = 0,1,\ldots,N-1$$

Ayrık Fourier Dönüşümleri

AFD:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn} \quad , \quad k = 0, 1, \cdots, N-1$$

Ters-AFD:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)kn} \quad , \quad n = 0, 1, \cdots, N-1$$